

504MAN Resource Management

**Enterprise Resource Planning (ERP)
& JIT/Lean Operations/Human
Resources and
Strategic Operations
Management
Week 5: Session 1 and Session 3**



Register...



Intended Module Learning Outcomes

1. Appraise the development of the human resource function.
2. Critically examine the traditional and contemporary approaches of recruitment and selection.
3. Analyse the procedures and practices of performance management and remedial activities.
- 4. Investigate and plan the resources required to achieve an organisation's aims and objectives.**
- 5. Evaluate and appraise the use of resources against organisational plans.**
6. Formulate plans to ensure that resources are maximised and remedial actions are taken where appropriate.

Session Outcomes

At the end of this session learners should be able to:



Understand Materials Requirements Planning (MRP) and Enterprise Resource Planning (ERP)

Recognize MRP and ERP roles as the source of the information necessary for planning and control

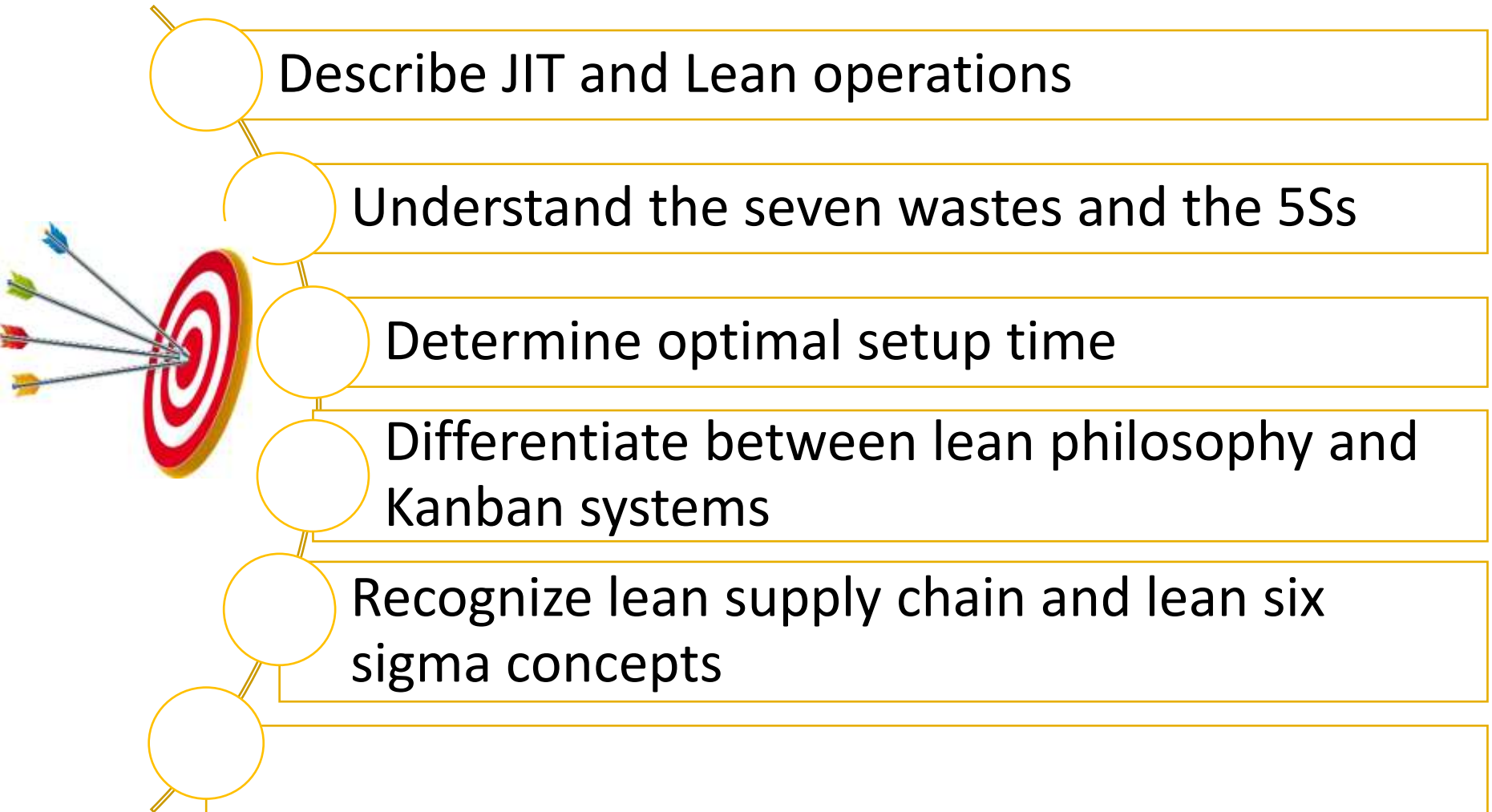
Explain the importance and difficulties associated with MRP and ERP systems

Place MRP and ERP in the context of operations planning and control



Session Outcomes

At the end of this session learners should be able to:



ERP
ENTERPRISE
RESOURCE
PLANNING

The image features the text 'ERP ENTERPRISE RESOURCE PLANNING' rendered in large, three-dimensional block letters. The letters are arranged in four rows. The top row contains 'ERP' in green. The second row contains 'ENTERPRISE' in white. The third row contains 'RESOURCE' in green. The bottom row contains 'PLANNING' in white. The letters have a slight shadow and are set against a dark gray background. The overall composition is a 3D perspective view of the text.

Introduction Video

- Kindly watch the video:
- <https://youtu.be/ZcYk2E6YCY0>



What is ERP ?

..E..

An Enterprise is a group of people with a common goal, which has certain resources at its disposal to achieve that goal.

..R..

Resources included are money, and all other things that are required to run the enterprise.

..P..

Planning is done to ensure that nothing goes wrong.



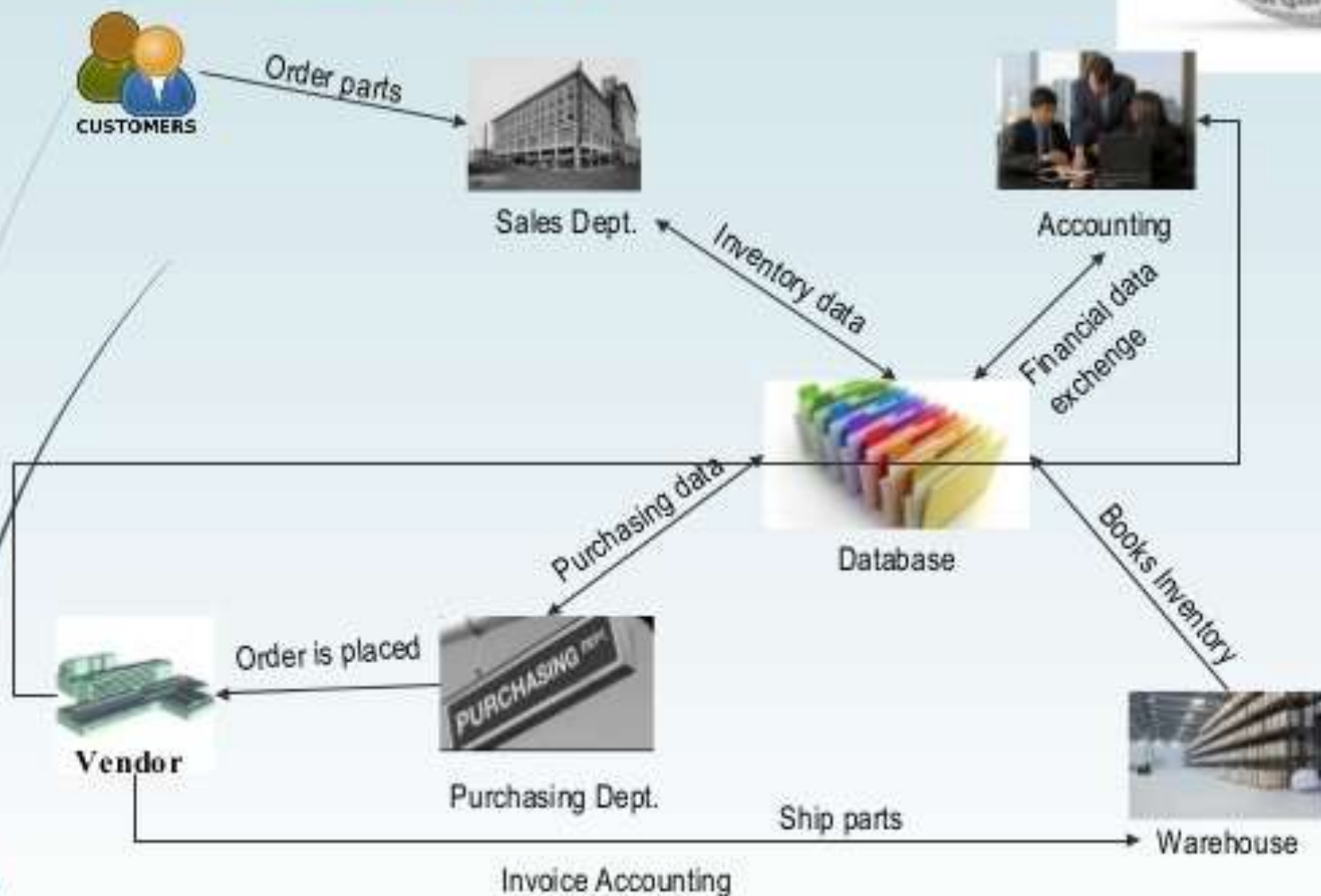
- ENTERPRISE RESOURCE PLANNING(ERP) is a cross-functional enterprise system driven by an integrated suite of software modules that supports the basic internal business processes of a company

What is ERP?



- The practice of consolidating an enterprise's planning, manufacturing, sales and marketing efforts into one management system.
- Combines all databases across departments into a single database that can be accessed by all employees.
- ERP automates the tasks involved in performing a business process.

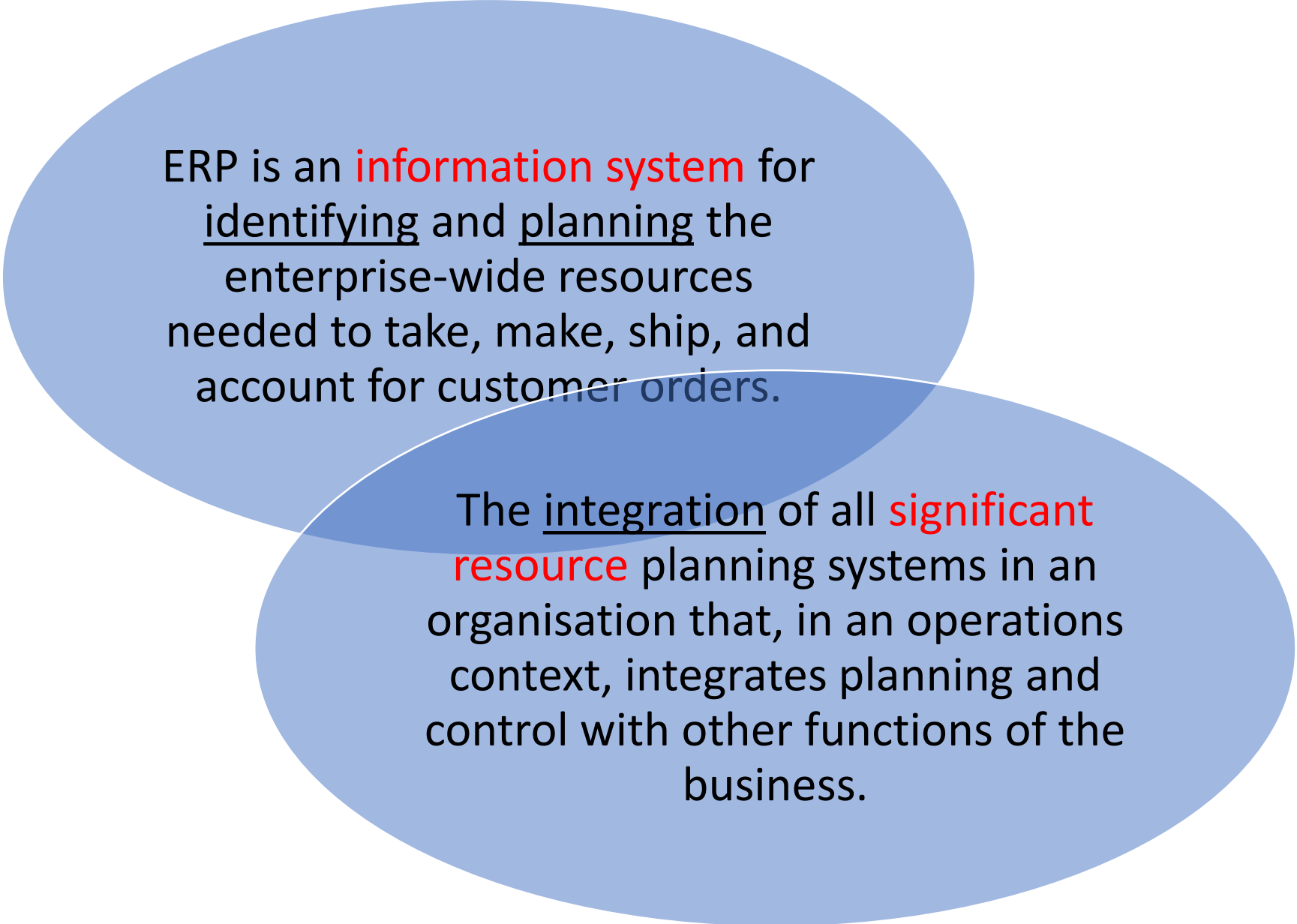
Centralized System : ERP example



In general ERP is :

“a business **strategy** and set of **industry-domain-specific** applications that build customer and shareholder **communities** value network system by enabling and optimizing enterprise and **inter-enterprise** collaborative operational and financial processes” (Source: Gartner’s Research Note SPA-12-0420)

.



ERP is an **information system** for identifying and planning the enterprise-wide resources needed to take, make, ship, and account for customer orders.

The integration of all **significant resource** planning systems in an organisation that, in an operations context, integrates planning and control with other functions of the business.

Goal of ERP

The goal of our enterprise is simply to get, keep, and grow customers.

GET

- **Acquire** profitable customers

KEEP

- **Retain** profitable customers longer.
- **Win back** profitable customers
- **Eliminate** unprofitable customers

GROW

- **Up sell** additional products in a solution
- **Cross sell** other products to customers
- Reduce **service and operations cost**.

Mini Case Study



Activity

20 minutes

OM in Action

Managing Benetton with ERP Software

Thanks to ERP, the Italian sportswear company Benetton can probably claim to have the world's fastest factory and the most efficient distribution in the garment industry. Located in Ponzano, Italy, Benetton makes and ships 50 million pieces of clothing each year. That is 30,000 boxes every day—boxes that must be filled with exactly the items ordered going to the correct store of the 5,000 Benetton outlets in 60 countries. This highly automated distribution center uses only 19 people. Without ERP, hundreds of people would be needed.

Here is how ERP software works:

1. *Ordering:* A salesperson in the south Boston store finds that she is running out of a best-selling blue sweater. Using a laptop PC, her local Benetton sales agent taps into the ERP sales module.
2. *Availability:* ERP's inventory software simultaneously forwards the order to the mainframe in Italy and finds that half the order can be filled immediately from the Italian warehouse. The rest will be manufactured and shipped in 4 weeks.

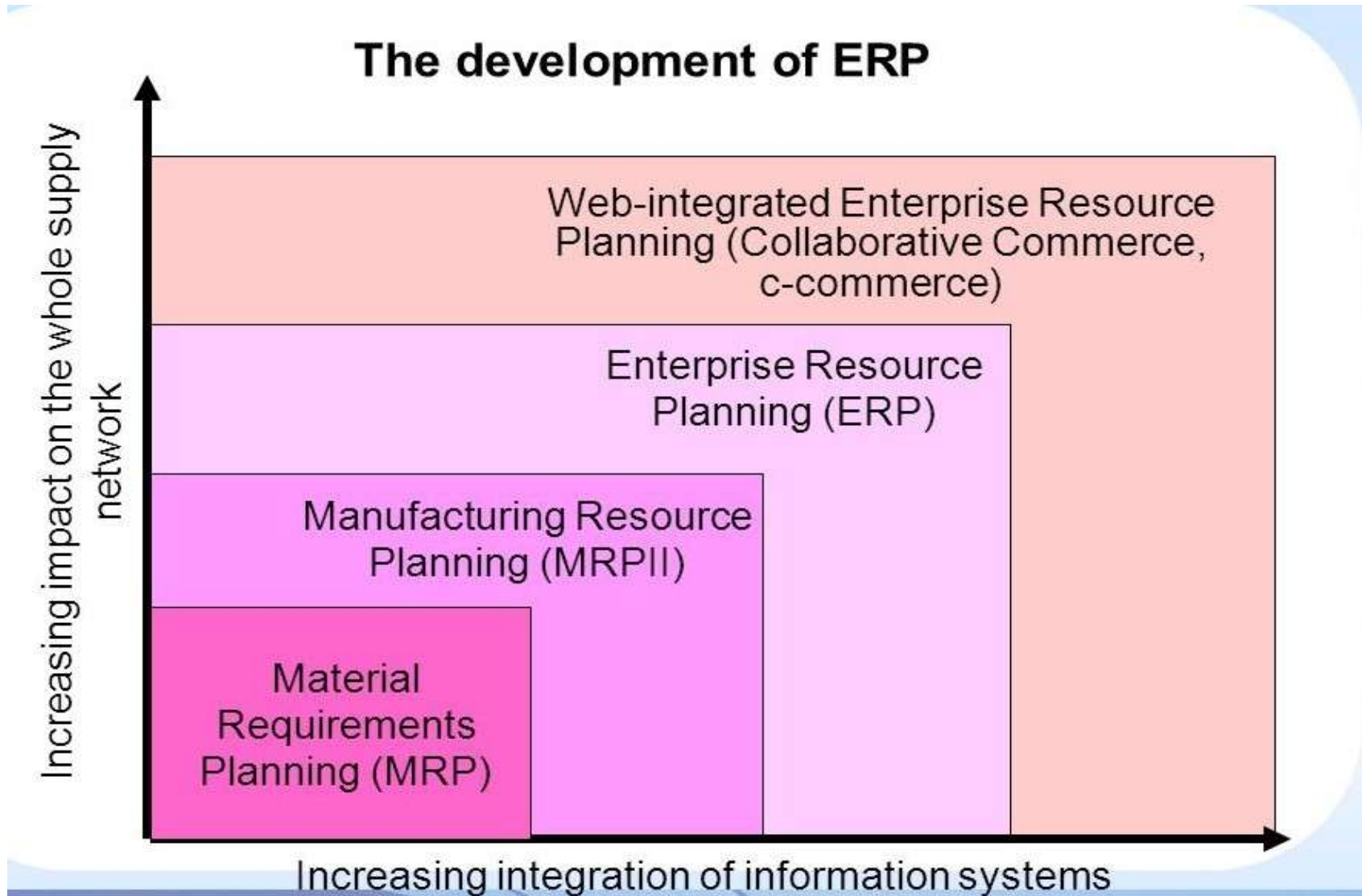
3. *Production:* Because the blue sweater was originally created by computer-aided design (CAD), ERP manufacturing software passes the specifications to a knitting machine. The knitting machine makes the sweaters.
4. *Warehousing:* The blue sweaters are boxed with a radio frequency ID (RFID) tag addressed to the Boston store and placed in one of the 300,000 slots in the Italian warehouse. A robot flies by, reading RFID tags, picks out any and all boxes ready for the Boston store, and loads them for shipment.
5. *Order tracking:* The Boston salesperson logs onto the ERP system through the Internet and sees that the sweater (and other items) are completed and being shipped.
6. *Planning:* Based on data from ERP's forecasting and financial modules, Benetton's chief buyer decides that blue sweaters are in high demand and quite profitable. She decides to add three new hues.

Sources: *Forbes* (December 2, 2011); *The Wall Street Journal* (April 10, 2007); *Information Week* (June 13, 2005); and *MIT Sloan Management Review* (Fall 2001).

Read the mini case study “Managing Benetton with ERP Software” and discuss them with your peers

(Slack and Brandon-Jones 2018)

Development of ERP



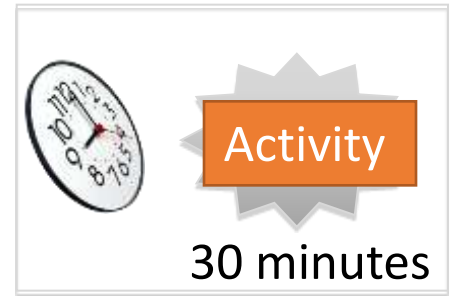
(Slack et al 2013)

Difficulty of implementation

- Very complex
- Extremely costly and time intensive
Typical: over \$10,000,000 and over a year to implement
- Company may implement only certain modules of entire ERP system
- You will need an outside consultant



Individual research & share

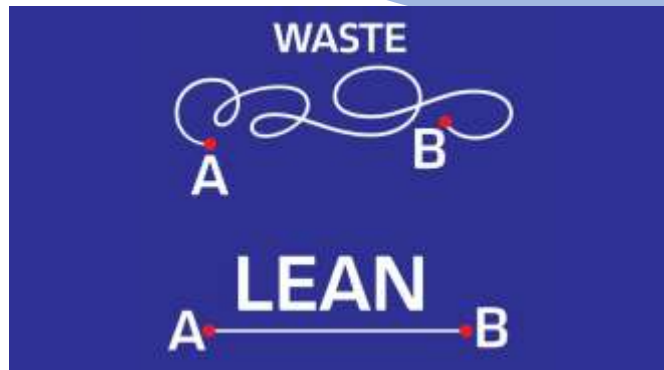


Undertake online research
and answer the question;

**“Are organisations in the
sector you are focusing on
using ERP? Provide examples
of these companies and how
they use ERP”.**

JIT and Lean Operations

Lean is a philosophy of production that emphasizes the **minimization** that the amount of all the resources used in the various activities of an enterprise.

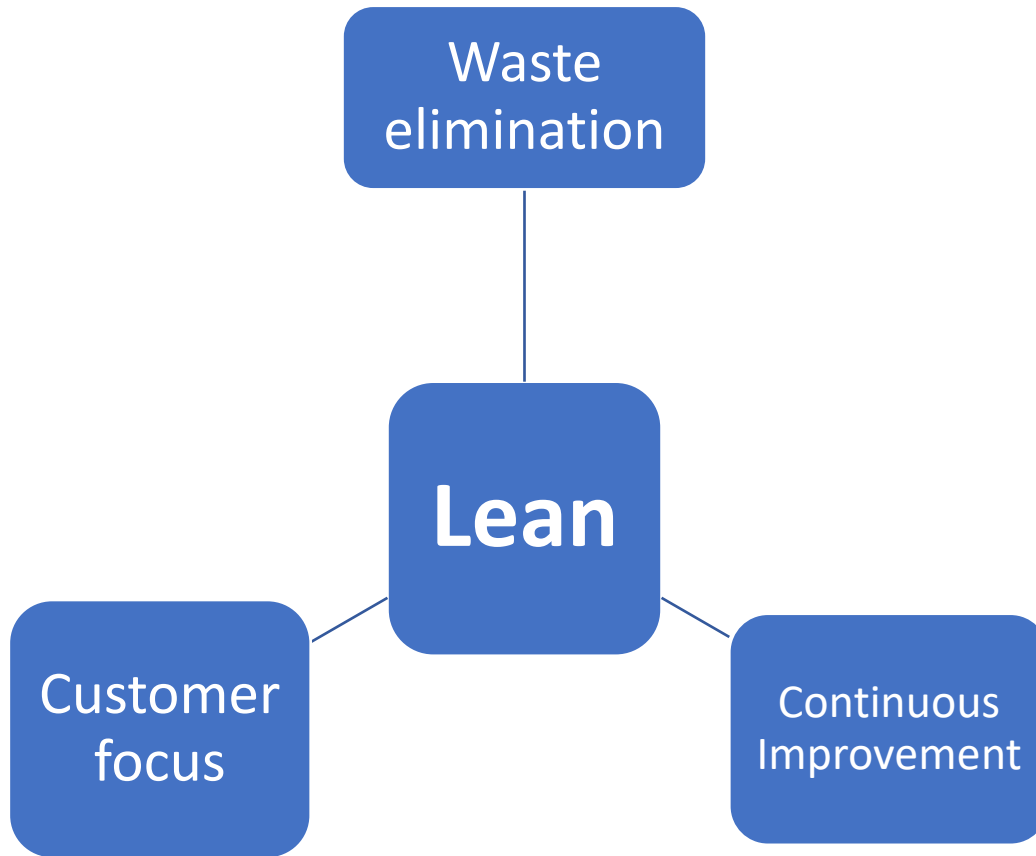


It involves identifying and **eliminating non-value adding** activities in design, production, supply chain management and dealing with customers.

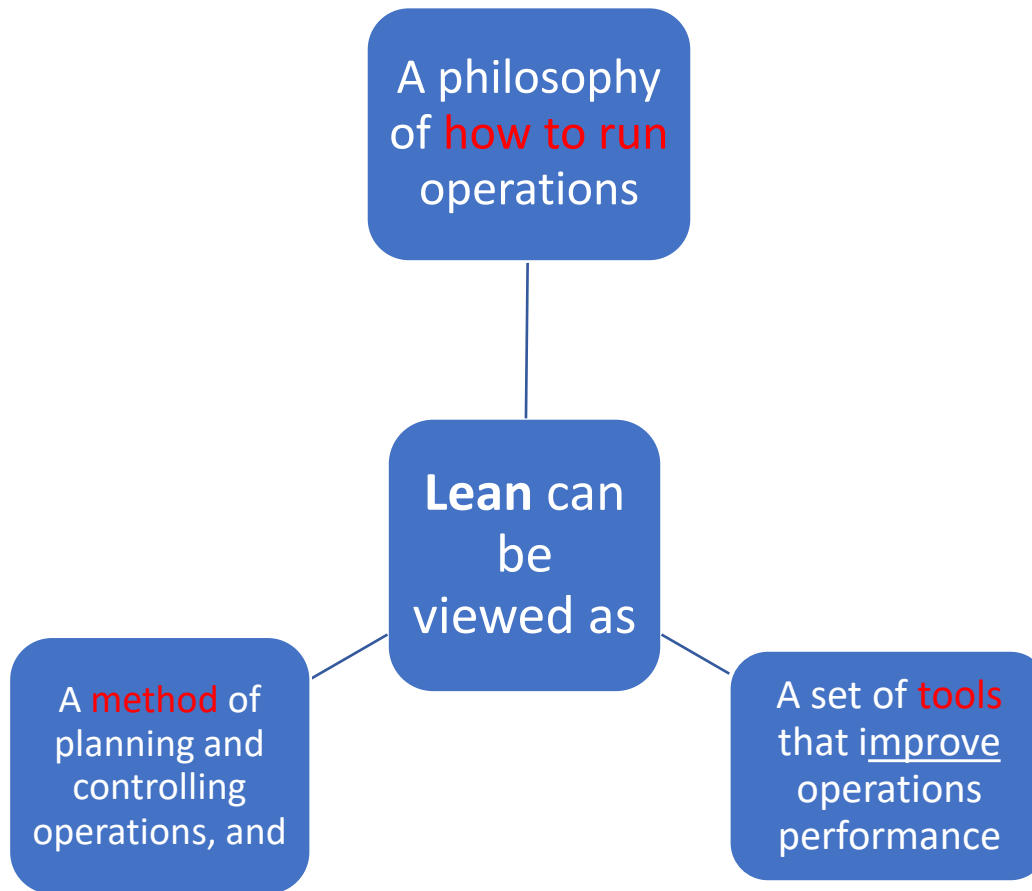
Lean operations supply the customer with **exactly** what the customer wants when the customer wants it, **without waste**, through continuous improvement

Driven by
“**pulling**”
customer
orders

Elements of lean



Three perspectives of lean



Just-in-time (JIT)

<https://youtu.be/cAUXHJBB5CM>

JIT is a philosophy of manufacturing based on planned **elimination of all waste** and continuous improvement of productivity

Used synonymously with “**Lean**”

It applies to all forms of manufacturing and many service industries as well

JIT and Competitive Advantage

JIT TECHNIQUES:

Suppliers:

Few vendors; Supportive supplier relationships;
Quality deliveries on time, directly to work areas.

Layout:

Work-cells; Group technology; Flexible machinery; Organized workplace; Reduced space for inventory.

Inventory:

Small lot sizes; Low setup time; Specialized parts bins

Scheduling:

Zero deviation from schedules; Level schedules;
Suppliers informed of schedules; Kanban techniques

Preventive maintenance:

Scheduled; Daily routine; Operator involvement

Quality production:

Statistical process control; Quality suppliers; Quality within the firm

Employee
empowerment:

Empowered and cross-trained employees; Training support;
Few job classifications to ensure flexibility of employees

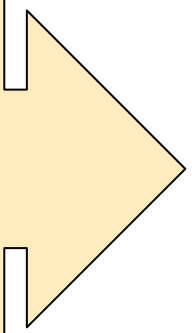
Commitment:

Support of management, employees, and suppliers

JIT can lead to Competitive Advantage – How?

WHICH RESULTS IN:

Quality improvement reduces waste
Cost reduction adds pricing flexibility
Rework reduction



WHICH WINS ORDERS BY:

- **Faster response** to the customer at;
 - **lower cost**;
 - & **higher quality** –
- A Competitive Advantage**

Taiichi Ohno's 7 wastes

Noted on the Toyota Production System



Eliminate Waste

- **Waste** is anything that does not add value from the customer point of view.
- Storage, delay, waiting in queues, and defective products;
- That do not add value and are 100% waste



Lean 5S in Traditional "housekeeping"

Sort/segregate

- when in doubt, throw it out

Simplify/straighten

- methods analysis tools

Shine/sweep

- clean daily

Standardize

- remove variations from processes

Sustain/self-discipline

- review work and recognize progress

Reading activity – 5 minutes

- Read the LinkedIn article on Aula to provide you with a snapshot of Lean 5S housekeeping method.

Building a Lean Organization

Lean systems tend to have the following attributes

Respect and develop employees

Empower employees

Develop worker flexibility

Develop collaborative partnerships with suppliers

Eliminate waste by performing only value-added activities



KAI

改

CHANGE

ZEN

善

GOOD

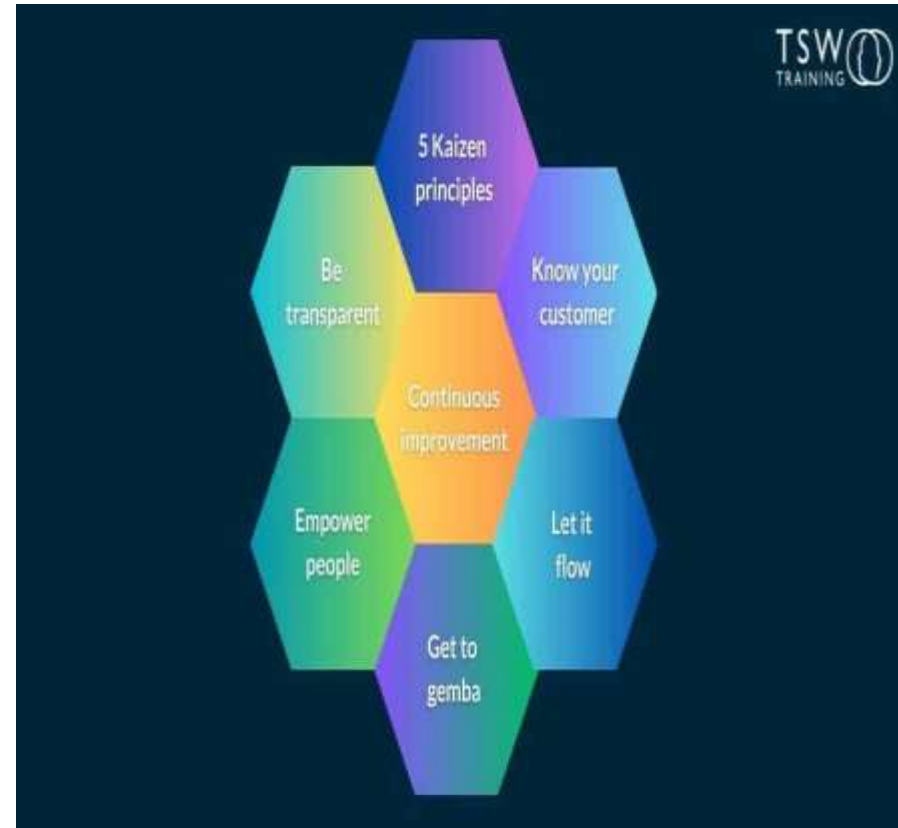
=

CONTINUAL
IMPROVEMENT

KAIZEN-Five Principles

(Masaaki Imai, 1986)

- **Know your customer:** Work out what **customers want and value** so you can improve their experience.
- **Let it flow:** Make **zero-waste** the goal & encourage your employees to reduce their waste.
- **Get to gemba:** “Gemba” means “the actual place” in Japanese, meaning you should go to **where value can be created** and take action
- **Empower people:** Make sure your teams have the **clarity, knowledge, equipment and systems** they need to succeed
- **Be transparent:** **Make improvements tangible by using data**, so your teams can see the impact they’re having.



What it means – KAIZEN

(Masaaki Imai, 1986)

- Kaizen is a Japanese term that means “**change for the better**” or “continuous improvement”.
- It's been adapted by organisations as a business strategy to help make **small changes** over time.
- Large-scale changes can make employees feel *unsure, worried and even disinterested* in a job.
- It involves to **take small steps** to *implement change* and make *continuous improvements*.
- The team have **time to adapt and contribute to these changes**, and it doesn't all happen at once.
- <https://youtu.be/Zc8DM7aQgIU>

Toyota Production System

- **Continuous improvement**
 - Build an organizational culture and value system that stresses improvement of all processes, **kaizen**
 - Part of everyone's job
- **Respect for people**
 - People are treated as knowledge workers
 - Engage mental and physical capabilities
 - Empower employees



Toyota Production System

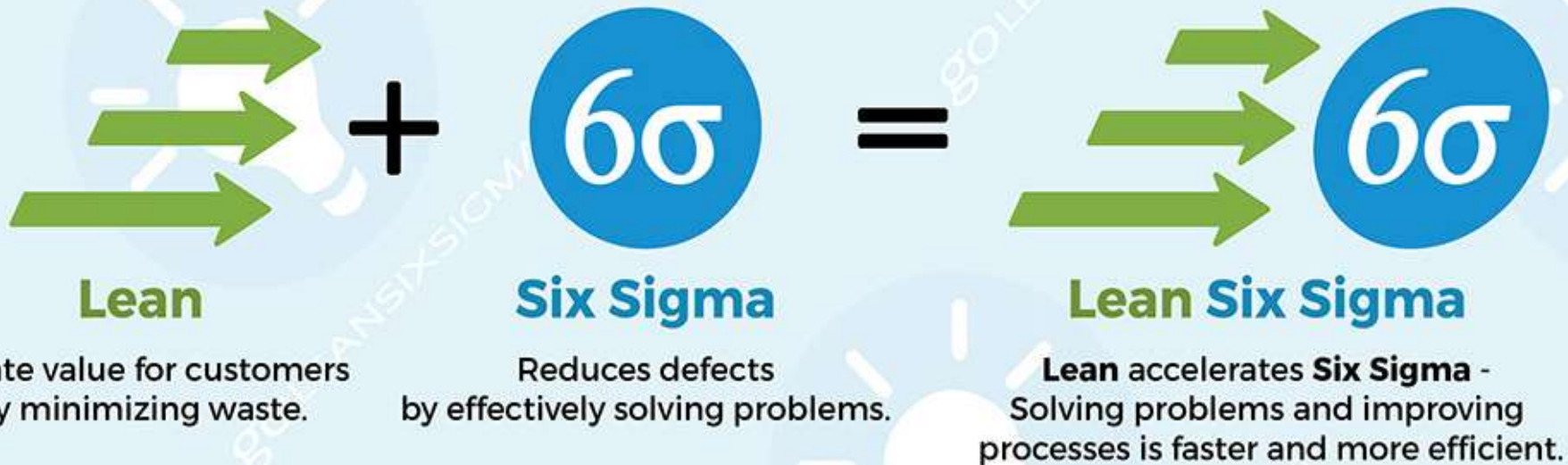
- Processes and standard work practice
 - Work shall be completely specified as to content, sequence, timing, and outcome.
- Internal and external customer-supplier connections are direct.
- Material and service flows must be simple and directly linked to the people or machinery involved.
- Process improvement must be made in accordance with the scientific method at the lowest possible level of the organization.

Lean Six Sigma



10 mins

What Is Lean Six Sigma?



Watch the video

https://youtu.be/p8VxP_a_S28

**Human Resources and
Strategic Operations
Management
Week 5: Session 3**



Intended Module Learning Outcomes

1. Appraise the development of the human resource function.
2. Critically examine the traditional and contemporary approaches of recruitment and selection.
3. Analyse the procedures and practices of performance management and remedial activities.
- 4. Investigate and plan the resources required to achieve an organisation's aims and objectives.**
- 5. Evaluate and appraise the use of resources against organisational plans.**
6. Formulate plans to ensure that resources are maximised and remedial actions are taken where appropriate.

Session Outcomes

At the end of this session learners should be able to:

- 1 Understand the strategic importance of human resource management
- 2 Appreciate the problems that firms have in managing human resource management
- 3 Understand the changes to human resources over time within operations management
- 4 Recognize the need for ongoing development, training and learning within organizations



‘The basis of
competitive
advantage is moving
from capital, natural
resources to **human
capital**’

(Tampoe 1994)

The **skills base** is one of the firm's main assets.

It is **hard** for competitors to imitate ... this calls for an attitude to encourage learning and to reward efforts which add to the firm's knowledge.

Skills **go out of date** and need constant replenishment. In the long term what is most important may not be the skills, but the ability to keep learning new ones (Grindley 1991)

- Human resources matter much more than they ever did.
- So one of the key skills in strategic operations management is going to be learning how to motivate and direct such resources.
- *Can you think of different ways of motivating individuals and team?*

A research on the global automobile industry in the 1980s showed that there were very significant performance differences between the best plants in the world (almost entirely Japanese operated at that time) and the rest.

The gaps were not trivial (Womack et al., 1990):

- “... our findings were eye-opening. The Japanese plants require one-half the effort of the American luxury-car plants, half the effort of the best European plant, a quarter of the effort of the average European plant, and one-sixth the effort of the worst European luxury-car producer. At the same time, the Japanese plant greatly exceeds the quality level of all plants except one in Europe – and this European plant required four times the effort of the Japanese plant to assemble a comparable product ...”
- Why do you think this is? Do you think national culture has anything to do with it?

What they found was that there were few differences in areas like automation – indeed, in many cases **non-Japanese plants had higher levels of automation and use of robots.**

But there were major differences in the **way work was organized** and in the approach taken to **human resources.**

Key elements in human resource management in strategic operations

Area	Key elements
Commitment to people as strategic resources	Employment security Choosing the right people Valuing and rewarding them Wage compression Symbolic egalitarianism
Shared purpose	Strategic leadership Shared planning processes Policy deployment Information sharing Employee ownership
Enabling structures	Appropriate organization design Job and work organization design Devolved decision making Supportive communications
Shared learning and development	Commitment to training and development Embedding a learning cycle Measurement Continuous improvement culture
Shared involvement	Teamworking Cross-boundary working Participation and involvement mechanisms Stakeholder focus and involvement



Activity

5 minutes

Traffic Light *it*

Where is your learning at?



Red = I have misunderstood some of the topics. (write down 2 questions that you need help with)

Amber = I have understood most of the topics so far (write down 1 question and 1 key theme you have understood)

Green = I have understood the topics so far (write down 2 key themes from the lesson to show your understanding)

Assignment support



Session Reference List

- Greasley, A. (2013) Operations Management. 3rd edn. John Wiley & Sons
- Heizer, J., Render, Barry, author, & Munson, Chuck, author. (2017). Principles of operations management sustainability and supply chain management (Tenth edition.; Global ed.).
- Hill, A. and Hill, T. (2018) Essential operations management. Second edition. London: Palgrave
- Slack, N., Brandon-Jones, Alistair, author, & Johnston, Robert, author. (2013). *Operations management* (Seventh ed.). Harlow, England: Pearson
- Slack, N. (2015) Operations and process management: principles and practice for strategic impact. Fourth edition. Harlow, England: Pearson Harlow, England: Pearson
- Slack, N. and Brandon-Jones, A. Johnston R. (2016) Operations Management. Eighth edition,
- Slack, N. and Brandon-Jones, A. (2018) Operations and process management: principles and practice for strategic impact. Fifth edition. Harlow, England: Pearson



Reading

Essential Reading

- Stevenson, W.J. (2021). *Operations Management* (fourteenth edition). McGraw Hill
- Torrington, D., Hall, L., Atkinson, C. (2020). *Human Resource Management* (11th edition). Pearson

Recommended Reading

- Noe, R.A., Hollenbeck, J.R., Gerhart, B.A, & Wright, P.M. (2022). *Fundamentals of human resource management* (Ninth edition). McGraw Hill.
- Noe, R.A., Hollenbeck, J.R., Gerhart, B.A, & Wright, P.M. (2022). *Human resource management: gaining a competitive advantage* (Thirteenth edition). McGraw Hill.
- Heizer, J., Render, B., & Munson, C. (2023). *Principles of Operations Management: Sustainability and Supply Chain Management* (Twelfth edition). Pearson.
- Slack, N., & Brandon-Jones, A. (2021). *Operations and process management* (Sixth edition). Pearson.
- Slack, N., & Brandon-Jones, A. (2018). *Operations and process management principles and practice for strategic impact* (Fifth edition). Pearson.
- Hill, A., & Hill, T. (2018). *Essential operations management* (Second edition). Palgrave.

Additional Reading

- Bratton, J., Gold, J., Bratton, A., (2012) *Human Resource Management: Theory and Practice*. 5th edn. Basingstoke: Palgrave MacMillan
- CMI Website : Management Direct